

ASTRATRACK — PERFORMANCE AUDIT REPORT

Run ID: RUN_1789091947 | Timestamp: 2026-09-11T01:59:07Z | SIH 2026

OVERALL EVALUATION VERDICT: FAIL

1. Mission Configuration & Pipeline

Scenario	01 — BASELINE	Detector	Classical CV (HSV)
Controller	PID_CONTROL	Filter	Kalman
Disturbance	NORMAL	Duration	2.63 s

2. Operational & Kinematic Metrics (Actual Measured Telemetry)

Metric Name	Measured Value	Units	Metric Name	Measured Value	Units
Avg Tracking Error	22.26	px	Max Tracking Error	178.1	px
RMS Tracking Error	33.8	px	Lock Retention Rate	99.4	%
Acquisition Time	0.03	s	Re-Acquisition Time	0.0	s
Target Loss Count	0	count	Recoveries Count	0	count
System Rate	11.2	FPS	Pipeline Latency	77.99	ms
Detector Inference	12.35	ms	Detection Conf	0.929	0-1

3. Quantitative Mission Criteria Evaluation

Criterion	Target Threshold	Measured Actual	Audit Status
Lock Retention Rate	>= 85.0%	99.4%	PASSED
RMS Tracking Error	<= 15.0 px	33.80 px	FAILED
Maximum Tracking Error	<= 40.0 px	178.10 px	FAILED
Processing Pipeline Latency	<= 35.0 ms	77.99 ms	FAILED
Mean Re-Acquisition Time	<= 2.00 s	0.00 s	PASSED

Integrity Guarantee: All displayed metrics are derived strictly from live simulation telemetry. No placeholder or synthesized figures were used in this audit.